

Advance Project Phase 4

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Abstract

This study explores advancements in sentiment analysis for mental health through the implementation of the PEFT with LoRA architecture on a pre-trained model across a multi-class classification task. Utilizing a dataset divided into training and evaluation sets, we enhance model performance by leveraging Parameter-efficient Fine-tuning (PEFT). The results from Phase 4, showing a significant improvement over Phase 3, emphasize the effectiveness of our approach with the highest accuracy reaching 84% .

Introduction

Mental health remains a critical area for research, particularly in the context of analyzing sentiment on social media platforms. The objective of this phase of the project was to apply advanced machine learning techniques, specifically leveraging PEFT with LoRA, to better categorize and understand the nuances of mental health-related discussions online. This work builds on our previous findings in Phase 3, aiming to refine accuracy and model responsiveness through sophisticated tuning methods.

Problem Domain Overview

Mental health challenges, including issues such as depression, anxiety, and stress, affect millions of people worldwide, posing a significant public health concern. The World Health Organization (WHO) estimates that roughly 1 in every 8 individuals globally experiences a mental health disorder, highlighting the critical need for tools that can help monitor and analyze trends in mental health [1]. Social media platforms have become a prevalent space for individuals to express their emotions and mental health struggles, providing a valuable source of data for understanding these issues on a large scale. However, analyzing such data is complex due to the nuanced and often ambiguous language people use to express their emotions and mental health experiences.

Sentiment analysis, a branch of natural language processing (NLP), plays a crucial role in interpreting and classifying emotional content within text data. Specifically for mental health, sentiment analysis can help identify patterns in public sentiment that indicate mental distress. This analysis is beneficial to mental health professionals, policymakers, and organizations by enabling more timely insights and responses. However, this task requires advanced techniques to capture the subtleties of language, including variations in tone, colloquialisms, sarcasm, and emotional intensity commonly found on social media [2].

Traditional approaches to sentiment analysis often involved rule-based systems and sentiment lexicons, but these methods tend to struggle with the intricacies of human language, which limits their effectiveness for mental health applications. Recent advancements in machine learning, especially in deep learning, have significantly improved the accuracy of sentiment analysis. Transformer-based models, like BERT and GPT, excel at understanding context and language patterns, making them especially useful in analyzing sentiments related to mental health. These models have shown high accuracy rates in detecting sentiment across various types of datasets, highlighting their effectiveness in capturing subtle emotions in text [3].

Artificial Intelligence Technologies in Mental Health Sentiment Analysis

In recent years, several AI-driven techniques have been applied to sentiment analysis, specifically in the context of mental health. Early machine learning models like Support Vector Machines (SVM) and Logistic

Regression set the foundation for sentiment classification. However, these methods are limited in their ability to capture context and long-term dependencies in language. Deep learning models, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, have improved accuracy in this field by capturing sequential patterns in text [4]. Despite their advantages, these models can still struggle with understanding complex language structures.

The development of transformer-based models, especially BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformers), has marked a significant advancement in sentiment analysis due to their powerful context-understanding capabilities. Studies employing BERT for analyzing mental health sentiment in social media posts have reported high accuracy rates, often exceeding 85%. For instance, BERT-based models have been shown to effectively identify subtle expressions of mental health-related emotions, such as distress and positivity [5] [6]. Another study, which used CNN and LSTM models for sentiment classification, reported high predictive accuracy, with some models achieving over 90%. These findings underscore the potential of advanced deep learning models to handle the complex language dynamics associated with mental health sentiment analysis, providing a valuable tool for detecting trends and distress signals in online discourse [7] [8].

Review of Studies Using the Dataset

In our review of existing literature on sentiment analysis applied to datasets similar to the "Sentiment Analysis for Mental Health," we found that transformer models, such as those based on BERT, are widely utilized due to their exceptional ability to handle nuanced emotional expressions. For instance, one study demonstrated that a BERT-based model achieved both 80% accuracy and an F1 score of 0.80 in analyzing social media texts related to mental health (Tamim et al., 2023) [9].

For the similar datasets researchers have examined platforms like Twitter, Reddit, and other social media sites to detect mental health markers such as anxiety, depression, and stress. One study that applied BERT models to Twitter posts related to mental health reported high accuracy and F1 scores, demonstrating the suitability of transformer-based models for analyzing sentiment in such data [10]. Another study focused on Reddit comments, using LSTM-based models, and showed that these models are reliable in distinguishing different types of mental health sentiments. These examples illustrate how similar datasets can provide valuable insights when combined with advanced AI techniques [11].

We have not found any studies that specifically used this dataset, and as per your requirement, if no previous studies have used this dataset specifically, it is beneficial to examine similar research in greater detail. So, we found this study very relative, analyzing studies on Twitter and Reddit data that utilize mental health sentiment analysis could offer insights into the approaches and performance metrics relevant to this dataset, as well as highlight limitations these studies encountered in capturing complex mental health expressions.

Hypothesis Statements

We hypothesize that by applying open-source machine learning software to the “Sentiment Analysis for Mental Health” dataset, it may be possible to develop useful technology for identifying and categorizing mental health-related sentiments. Such a tool could provide critical insights into public mental health trends, potentially aiding mental health professionals and policymakers in making timely interventions.

Additional hypotheses aligned with the project’s objectives include:

- 1.** Transformer-based models, like BERT, will achieve better performance in accurately classifying mental health-related sentiment on social media than traditional machine learning methods.
- 2.** The PEFT with LoRA technique will significantly outperform the baseline DistilBERT model in terms of accuracy and other evaluation metrics for mental health sentiment analysis on NLP-based datasets.

1 Proof: The hypothesis is validated by the results. While the use of embeddings without fine-tuning provides moderate improvements, the fine-tuning of transformer-based models like DistilBERT leads to significantly better performance, with accuracy and F1-scores consistently surpassing traditional methods. This highlights the necessity of fine-tuning for achieving optimal results in mental health sentiment analysis tasks.

2 proof: PEFT with LoRA technique achieved an accuracy of 0.840, significantly outperforming the baseline DistilBERT’s accuracy of 0.808.

Materials and Methods

Dataset Description:

The dataset utilized in this study was sourced from a Kaggle dataset titled 'Sentiment Analysis for Mental Health', compiled by Suchintika Sarkar. This dataset was assembled to facilitate research in the sentiment analysis of statements related to mental health conditions. It comprises 52,681 textual statements categorized into seven mental health statuses: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. These categories help differentiate the emotional context or mental health status expressed in each statement.

The primary objective of the dataset is to provide a resource for training machine learning models that can recognize and classify various mental health conditions from text data, making it a valuable tool for researchers and clinicians in the field of mental health. The data was collected to reflect a broad spectrum of emotions and conditions, providing a comprehensive view of public sentiment related to mental health issues.

Category	Count	Percentage
Normal	16343	31%
Depression	15404	29%
Suicidal	10652	20%
Anxiety	3841	7%
Bipolar	2777	5%
Stress	2587	5%
Personality Disorder	1077	2%

Engineered Feature Statistics by Status:

This table summarizes the engineered feature statistics calculated for each status category in the dataset. The following features were extracted from the textual data:

- 1) Word Count: The total number of words in each statement.
- 2) Character Count: The total number of characters in each statement

3) Average Word Length: The average length of words in each statement.

We calculated the average word length by dividing the character count by the word count. The mean (average) and standard deviation of these features were computed for each mental health category (status). This analysis helps us to understand the distribution of text-based features across categories and provides a quantitative summary for further modeling.

Status	Word_count (Mean)	Word_count (Standard Deviation)	Char_count (Mean)	Char_count (Standard Deviation)	Avg_word_length (Mean)	Avg_word_length (Standard Deviation)
Anxiety	142.11	152.50	755.47	811.73	5.36	0.66
Bipolar	170.11	176.34	913.30	951.62	5.33	0.63
Depression	168.02	188.23	844.03	953.00	5.08	2.41
Normal	17.24	22.77	90.20	120.88	5.30	1.60
Personality disorder	160.93	217.02	858.26	1126.84	5.28	1.26
Stress	111.10	106.08	594.79	571.05	5.51	1.99
Suicidal	146.43	186.96	734.90	982.19	5.31	3.62

Target Variable Description:

The target variable for the study is the 'status' column, which is a multiclass classification task. The objective is to predict the mental health status (Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, Personality Disorder) based on the textual content of the 'statement' column.

Machine Learning:

In Phase 3

The process begins with data preprocessing, including handling missing data by dropping rows with missing values. Text cleaning removes URLs, HTML tags, special characters, numbers, and extra spaces. Stopwords are excluded using NLTK. Columns are renamed and reordered as "label" and "text," with categorical labels mapped to integers. The dataset is then split into training (A1) 80 %, evaluation (A2) 10 %, and test sets (B) 10%. A pre-trained DistilBERT tokenizer is applied for consistent padding and truncation.

The DistilBERT model is fine-tuned using Hugging Face's Trainer for classification. The fine-tuned model computes sentence embeddings for the test dataset by averaging the last hidden states. Finally, embeddings, along with labels and text, are saved in Parquet and CSV formats for further analysis with DF-Analyze

In phase 4

Data Preparation

The raw data were preprocessed to ensure quality and consistency for machine learning applications. The preprocessing steps included:

- **Tokenization:** Using a pre-trained tokenizer, each textual statement was converted into a sequence of tokens. This process involved breaking down paragraphs into sentences and sentences into words, which were then transformed into token IDs that could be processed by neural network models.
- **Cleaning:** The text data were cleaned to remove noise that could interfere with the models' training. This included stripping unnecessary punctuation, removing special characters, and correcting common spelling errors.
- **Splitting:** The dataset was randomly divided into two subsets: Subset A (50% of the data) for training and Subset B (50% of the data) for testing and validation, ensuring a representative distribution of classes in both subsets.

Model Selection and Configuration

The core of the analysis in Phase 4 centered around the implementation of the LoRA (Low-Rank Adaptation) technique integrated into a pre-trained language model. The models selected for comparison included:

- **Base Models:** Logistic Regression (LR), Stochastic Gradient Descent (SGD), Light Gradient Boosting Machine (LGBM), Random Forest (RF), and K-Nearest Neighbors (KNN).
- **Pre-trained Model:** A pre-fine-tuned DistilBERT model was used as the foundation for further tuning. DistilBERT, known for its efficiency and effectiveness in processing language data, provided a robust baseline for sentiment analysis.

LoRA Implementation

LoRA allows efficient fine-tuning of large pre-trained models by introducing low-rank learned matrices that adapt the pre-existing weights rather than altering them extensively. This method is particularly effective in preventing overfitting when dealing with smaller datasets. The key configurations for LoRA included:

- **Rank Selection:** The rank of the adaptation matrices was chosen based on preliminary experiments to balance performance and computational efficiency.

- **Training Parameters:** Learning rates, batch sizes, and training epochs were optimized through a series of grid searches to find the best settings for this specific dataset and task.

Training and Validation

- **Training:** The models were trained on Subset A using cross-entropy loss as the objective function to handle the multi-class classification nature of the task. The training process was monitored using a validation split from Subset A to tune hyperparameters and prevent overfitting.
- **PEFT (Parameter-Efficient Fine-tuning):** After initial training, PEFT was applied using LoRA to refine the models' performance further. This stage focused on fine-tuning the top layers of the DistilBERT model, where the most task-specific adaptations occur.

Performance Evaluation

- **Metrics:** The models' performance was evaluated using several metrics, including Accuracy, AUROC (Area Under the Receiver Operating Characteristic curve), Balanced Accuracy, F1 Score, NPV (Negative Predictive Value), PPV (Positive Predictive Value), Sensitivity, and Specificity.
- **Embeddings Extraction:** For Subset B, embeddings were extracted post-PEFT to analyze the impact of fine-tuning on the representation of text data related to different mental health conditions.

Discussion

Statistical Analysis

Interesting Findings

The most notable outcome of Phase 4 is the successful application of LoRA (Low-Rank Adaptation) to fine-tune a pre-trained DistilBERT model for the task of sentiment analysis in mental health contexts. The incorporation of PEFT (Parameter-Efficient Fine-tuning) and LoRA notably improved model performance across several key metrics:

- Accuracy increased significantly from the highest in Phase 3 (0.808) to 0.840 in Phase 4.
- Balanced Accuracy also saw a meaningful increase, indicating an enhanced ability of the model to perform well across various classes, not just the dominant ones.

This improvement is particularly significant as it demonstrates the effectiveness of LoRA in adapting large pre-trained models to specific tasks without extensive computational resources, aligning with the growing need for efficient AI solutions in practical applications.

Hypothesis Testing

The hypotheses posited at the outset of this research focused on whether advanced, parameter-efficient techniques like LoRA could outperform traditional methods in the multi-class classification of mental health sentiments. The results validate these hypotheses, with LoRA-enhanced models showing superior performance compared to baseline models used in Phase 3. This supports the theory that efficient fine-tuning methods can unlock higher potential in pre-trained models for specialized tasks without the need for extensive retraining.

Literature Comparison

Comparison with Phase 3

When comparing these findings to the results obtained in Phase 3, which used more conventional training methods, the advantages of integrating PEFT with LoRA are clear. The enhanced models not only performed better statistically but also showed improved generalization capabilities over unseen data, a critical aspect for real-world applications where model robustness is essential.

Comparison with External Studies

The results from Phase 4 align well with current literature suggesting that LoRA and similar methods provide substantial benefits over traditional fine-tuning techniques. Studies such as those by [Yang, S. K. G., & Contributors. (2023)] have reported improvements in performance metrics similar to those observed in our study when applying LoRA to other NLP tasks [11]. However, our application within the mental health sentiment analysis domain provides new insights into the versatility and potential of these approaches.

Discrepancies in performance, particularly in comparison to studies that did not utilize parameter-efficient techniques, can largely be attributed to the enhanced ability of LoRA to maintain the integrity of the pre-trained model's knowledge while effectively adapting to new tasks. This suggests that much of the traditional fine-tuning might dilute pre-trained models' generalized capabilities, which LoRA techniques manage to preserve.

Implications

The findings from this study have significant implications for future research and practical applications:

- For Researchers: The demonstrated effectiveness of LoRA suggests that similar parameter-efficient techniques should be explored and adapted for other complex NLP tasks beyond sentiment analysis.
- For Practitioners: The improved accuracy and efficiency of the models suggest that deploying such AI models in real-world settings, such as for monitoring social media for mental health issues, is becoming increasingly feasible.

Limitations and Future Work

While the results are promising, the study's limitations must be acknowledged. The dataset, sourced from Kaggle and inherently noisy due to its social media origin, may contain biases and inaccuracies in sentiment labeling. Future studies could enhance the dataset's reliability through rigorous validation or by incorporating expert annotations.

Moreover, further research could explore the application of LoRA in conjunction with other novel NLP techniques, such as transformer-based models beyond DistilBERT, to ascertain if these findings can be generalized across different architectures and larger datasets.

Results

Overview

In Phase 4, the project introduced the implementation of Parameter-Efficient Fine-tuning (PEFT) using the LoRA (Low-Rank Adaptation) architecture integrated into a pre-trained model for multiclass classification tasks. This phase focused on comparing the efficacy of PEFT against traditional training methods (Phase 3), specifically evaluating the models' ability to accurately classify mental health states from social media text.

Performance Evaluation

The results are based on comprehensive testing using both a holdout set and 5-fold cross-validation to ensure robust evaluation. The performance metrics considered were accuracy, area under the receiver operating characteristic curve (AUROC), balanced accuracy, F1 score, negative predictive value (NPV), positive predictive value (PPV), sensitivity, and specificity.

Holdout Set Performance

The table below summarizes the performance of various models on the holdout set:

Model	Selection	Embed Selector	ACC	AUR OC	Bal-ACC	F1	NPV	PPV	Sens	Spec
lr	embed_lgbm	lgbm	0.84	0.977	0.822	0.828	0.97	0.836	0.822	0.97
lr	embed_linear	linear	0.839	0.977	0.824	0.828	0.97	0.833	0.824	0.97
lr	none	none	0.839	0.977	0.821	0.83	0.97	0.841	0.821	0.97
lgbm	pred	none	0.839	0.971	0.825	0.831	0.97	0.838	0.825	0.97
lr	assoc	none	0.839	0.977	0.823	0.827	0.97	0.832	0.823	0.97
sgd	assoc	none	0.838	0.964	0.823	0.827	0.97	0.83	0.823	0.97
lgbm	assoc	none	0.838	0.972	0.826	0.829	0.97	0.833	0.826	0.97
lr	pred	none	0.838	0.977	0.823	0.826	0.97	0.83	0.823	0.97
sgd	none	none	0.838	0.964	0.823	0.826	0.97	0.83	0.823	0.97
sgd	pred	none	0.838	0.964	0.823	0.826	0.97	0.83	0.823	0.97
sgd	embed_linear	linear	0.837	0.976	0.823	0.826	0.97	0.83	0.823	0.97
sgd	embed_lgbm	lgbm	0.837	0.964	0.823	0.826	0.97	0.829	0.823	0.97
lgbm	none	none	0.837	0.972	0.825	0.829	0.97	0.832	0.825	0.97
lgbm	embed_lgbm	lgbm	0.837	0.972	0.827	0.828	0.97	0.83	0.827	0.97
lgbm	embed_linear	linear	0.837	0.972	0.826	0.827	0.97	0.829	0.826	0.97
rf	embed_lgbm	lgbm	0.835	0.953	0.83	0.823	0.969	0.817	0.83	0.97
rf	assoc	none	0.834	0.954	0.836	0.824	0.969	0.815	0.836	0.97
rf	none	none	0.834	0.951	0.839	0.82	0.969	0.805	0.839	0.97
rf	embed_linear	linear	0.834	0.953	0.835	0.822	0.969	0.812	0.835	0.97
rf	pred	none	0.833	0.954	0.837	0.819	0.969	0.804	0.837	0.97
knn	pred	none	0.827	0.938	0.816	0.82	0.968	0.825	0.816	0.968
knn	assoc	none	0.826	0.938	0.816	0.82	0.968	0.825	0.816	0.968
knn	embed_linear	linear	0.826	0.938	0.816	0.82	0.968	0.825	0.816	0.968
knn	none	none	0.826	0.938	0.816	0.82	0.968	0.825	0.816	0.968
knn	embed_lgbm	lgbm	0.82	0.938	0.81	0.82	0.96	0.82	0.816	0.96

			6		6		8	6		8
lr	wrap	none	0.82 2	0.966	0.80 6	0.81 3	0.96 7	0.82 3	0.806	0.96 7
sgd	wrap	none	0.82	0.961	0.80 2	0.81	0.96 7	0.82	0.802	0.96 6
lgbm	wrap	none	0.81 7	0.97	0.80 1	0.80 6	0.96 6	0.81 1	0.801	0.96 6
rf	wrap	none	0.81 6	0.948	0.81 4	0.80 1	0.96 6	0.79 1	0.814	0.96 6
knn	wrap	none	0.80 5	0.928	0.79 3	0.79 6	0.96 4	0.8	0.793	0.96 4
dummm y	embed_lgbm	lgbm	0.31 4	0.5	0.14 3	0.06 8	0.88 6	0.31 4	0.143	0.85 7
dummm y	assoc	none	0.31 4	0.5	0.14 3	0.06 8	0.88 6	0.31 4	0.143	0.85 7
dummm y	none	none	0.31 4	0.5	0.14 3	0.06 8	0.88 6	0.31 4	0.143	0.85 7
dummm y	embed_linea r	linear	0.31 4	0.5	0.14 3	0.06 8	0.88 6	0.31 4	0.143	0.85 7
dummm y	pred	none	0.31 4	0.5	0.14 3	0.06 8	0.88 6	0.31 4	0.143	0.85 7
dummm y	wrap	none	0.31 4	0.5	0.14 3	0.06 8	0.88 6	0.31 4	0.143	0.85 7

- **logistic Regression (LR)** with LightGBM embeddings (embed_lgbm) achieved the highest accuracy of 0.840 and an AUROC of 0.977, indicating excellent model performance and reliability in prediction.
- **Stochastic Gradient Descent (SGD)** and Random Forest (RF) models also showed strong performance, particularly in maintaining high specificity and NPV across different configurations.

5-Fold Cross-Validation Performance

This method was used to further validate the stability and reliability of the models across different subsets of the data. The results demonstrated consistent performance, reinforcing the findings from the holdout set evaluation.

5-Fold Performance on Holdout Set Table

model	selection	embed_select or	acc	auro c	bal- acc	f1	npv	ppv	sens	spec
lr	assoc	none	0.84	0.977	0.829	0.83 2	0.97	0.83 8	0.829	0.97
lr	embed_lgbm	lgbm	0.84	0.977	0.824	0.83 1	0.97	0.84	0.824	0.97
lr	pred	none	0.83 9	0.977	0.825	0.83	0.97	0.83 6	0.825	0.97
lr	none	none	0.83 9	0.976	0.821	0.83	0.97	0.84 1	0.821	0.97
lr	embed_linea r	linear	0.83 9	0.977	0.824	0.83	0.97	0.83 8	0.824	0.97
sgd	embed_linea r	linear	0.83 9	0.976	0.824	0.82 9	0.97	0.83 7	0.824	0.97
sgd	embed_lgbm	lgbm	0.83 9	0.975	0.824	0.82 9	0.97	0.83 7	0.824	0.97
sgd	assoc	none	0.83 9	0.976	0.823	0.82 8	0.97	0.83 5	0.823	0.97
lgbm	embed_linea r	linear	0.83 9	0.975	0.825	0.83 1	0.97	0.83 8	0.825	0.97
sgd	none	none	0.83 8	0.976	0.823	0.82 8	0.97	0.83 4	0.823	0.97
sgd	pred	none	0.83 8	0.967	0.823	0.82 8	0.97	0.83 4	0.823	0.97
lgbm	assoc	none	0.83 8	0.975	0.821	0.82 8	0.97	0.83 8	0.821	0.97
lgbm	pred	none	0.83 7	0.973	0.825	0.83 2	0.97	0.84 1	0.825	0.97
rf	embed_linea r	linear	0.83 7	0.955	0.833	0.82 5	0.97	0.82	0.833	0.97
lgbm	embed_lgbm	lgbm	0.83 6	0.975	0.823	0.82 6	0.97	0.83	0.823	0.97
rf	assoc	none	0.83 6	0.955	0.834	0.82 2	0.96 9	0.81 3	0.834	0.97
rf	embed_lgbm	lgbm	0.83 5	0.958	0.83	0.82 3	0.96 9	0.81 8	0.83	0.97
lgbm	none	none	0.83 5	0.975	0.821	0.82 7	0.96 9	0.83 6	0.821	0.96 9
rf	pred	none	0.83 3	0.953	0.837	0.81 6	0.96 9	0.8	0.837	0.97
rf	none	none	0.83 1	0.954	0.836	0.81 1	0.96 9	0.79 3	0.836	0.97
knn	assoc	none	0.82	0.94	0.807	0.81 4	0.96 7	0.82 4	0.807	0.96 7
knn	embed_linea r	linear	0.82	0.94	0.807	0.81 4	0.96 7	0.82 4	0.807	0.96 7
knn	none	none	0.82	0.94	0.807	0.81 4	0.96 7	0.82 4	0.807	0.96 7
knn	embed_lgbm	lgbm	0.81 9	0.94	0.807	0.81 3	0.96 7	0.82 3	0.807	0.96 7

knn	pred	none	0.819	0.94	0.806	0.813	0.966	0.823	0.806	0.966
lgbm	wrap	none	0.814	0.965	0.772	0.783	0.966	0.798	0.772	0.965
rf	wrap	none	0.806	0.942	0.785	0.765	0.964	0.751	0.785	0.965
lr	wrap	none	0.804	0.956	0.701	0.692	0.964	0.8	0.701	0.963
knn	wrap	none	0.793	0.917	0.741	0.749	0.962	0.769	0.741	0.962
sgd	wrap	none	0.793	0.934	0.687	0.68	0.962	0.784	0.687	0.961
dummy	embed_lgbm	lgbm	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	assoc	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	none	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	embed_linear	linear	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	pred	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	wrap	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857

- The models maintained high levels of accuracy and balanced accuracy, indicating effective generalization across the dataset.
- The F1 scores and other metrics remained robust, suggesting that the models are well-tuned and can handle varied data distributions effectively.

Discussion of Results

The implementation of PEFT with LoRA significantly enhanced the performance metrics compared to Phase 3 results, where Distil-BERT fine-tuning methods were employed. This enhancement can be attributed to LoRA's ability to efficiently adapt pre-trained model parameters to the specific nuances of the dataset used, without the need for extensive re-training.

- **Accuracy and AUROC improvements** reflect the models' enhanced capability to discriminate between different mental health conditions accurately.
- **Balanced accuracy** across models suggests that the tuning method effectively addressed class imbalance, a common challenge in mental health datasets.

Conclusion

The advanced tuning techniques applied in Phase 4 have clearly demonstrated their value in improving the precision and generalization of sentiment analysis models for mental health. These results not only validate the effectiveness of PEFT with LoRA but also suggest their potential applicability in other domains requiring nuanced text analysis.

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